

ROC Curves

DS 301

Iowa State University

- For logistic regression, LDA, and QDA, and the goal is to estimate this probability: $\hat{P}(Y = 1|X)$.
- We can set the threshold to be 0.5 such that $\hat{P}(Y = 1|X) > 0.5$, we label the observation a '1'.
- This will minimize the overall misclassification error rate.
- However, as we've seen, in some applications minimizing other types of errors may be more meaningful.

Varying the classifier threshold changes its true positive false positive rate.

ROC Curve

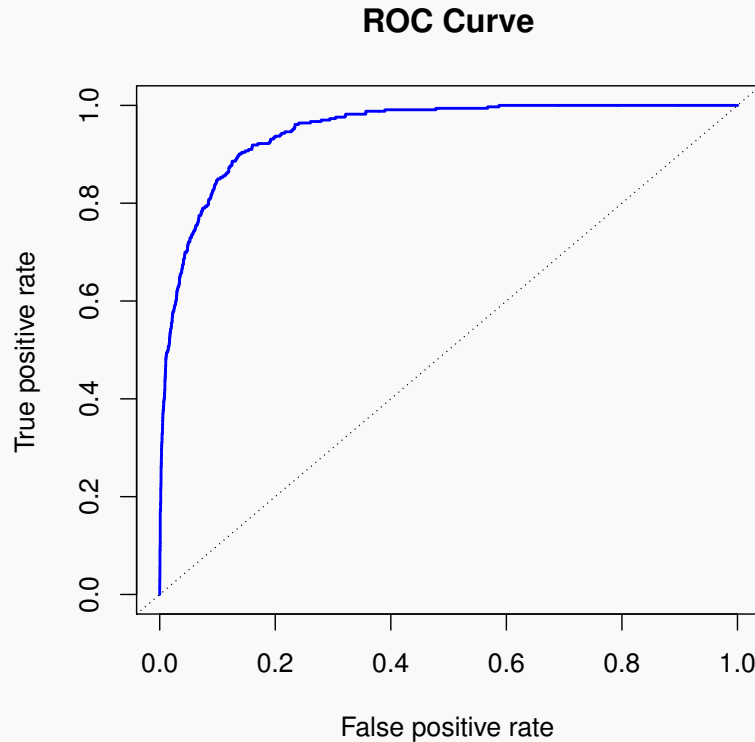


Figure 1: ROC curve is a graphic for simultaneously displaying the two types of errors for all possible thresholds. ROC curve for the LDA classifier on the Default data. Ideal ROC curve hugs the top left corner: indicates a high true positive rate and a low false positive rate

- across all possible thresholds The overall performance of a classifier, summarized over all possible thresholds, is given by the Area Under the Curve (AUC).
- We expect a classifier that performs no better than chance to have an AUC of 0.5.
- An $AUC > 0.8$ is considered quite good.

See R script: `roc_curve.R`